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3.2:

ASSISTING IN CAST APPLICATION AND PROVIDING CAST CARE

CAST/POP

- **Cast:** is an immobilizing device made up of layers of plaster or fiberglass bandages molded to the body part that it incases

Purpose

- To immobilize and hold bone fragments in reduction
- To apply uniform compression of soft tissues
- To permit early mobilization
- To correct and prevent deformities
- To support and stabilize weak joints

Types of Casts

a. Short arm cast

- Extends from below the elbow to the proximal palmar crease.

b. Gauntlet cast

- Extends from below the elbow to the proximal palmar crease, including the thumb (thumb)

c. Long arm cast

- Extends from upper level of axillary fold to proximal palmar crease elbow usually immobilized at right angle

d. Short leg cast: Extends from below knee to base of toes .

Types of casts Cont'd..

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e. Long-leg cast

- Extends from upper thigh to the base of toes foot is at right angle in neutral position.

f. Body cast or spica cast

- Encircles the trunk stabilizing the spine
- Casts that encase the body.

g. Spica case

- Incorporate the trunk and extremity

h. Shoulder spica cast

- A body jacket that encloses trunk, shoulder and elbow

Types of casts Cont'd..

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i. Hip spica

- Encloses trunk and al lower extremity

j. Single hip spica

- Extends from nipple line on to include pelvic and extends to include pelvis and one thigh.

k. Double hip spica

- Extends from hipple line or upper abdomen to include pelvis and extends to include both thighs and lower legs.

l. One and half hip spica

- Extends form upper abdomen includes one entire legs, and extends to the knee of the other.

Types of casts Cont'd..

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m. Cast-brace

- Is external support about a fracture that is constructed with hinges to permit early motion of joints
- Early mobilization and independence.
- It will promote the formation of bone man to
- It is applied after initial edema and pain have subsided and there is evidence of # stability.

n. Cyclinder case

- Can be used for upper or lower extremity.
- Used for fracture or dislocation of knee (lower extremity) or elbow dislocation (upper extremity)

Complication of casts

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Pressure of cast on neurovascular and bony structure causes:

- ☐ **Pressure sores**
- ☐ **Nerve palsies**
- ☐ **Compartment syndrome**
 - Trauma or surgery affecting an extremity will produce swelling w/c result of hemorrhage from bone and surrounding tissue and tissue edema.
 - Vascular insufficiency, nerve insufficiency and muscle compression due to unrelieving swelling can cause irreversible damage to an extremity

Comps of cast Cont'd...

8 = Immobility and confinement in a cast, particularly in body cast can result in multisystem problems.

- Nausea, vomiting and abdominal distension ass'ted with cast syndrome.
- Superior mesenteric artery syndrome, resulting in diminished blood flow to bowel.
- Paralytic illus
- Possible intestinal obstruction
- Acute anxiety rxn symptoms i.e. behavioral changes and autonomic responses with confinement in a space
- Increase RR,HR,BP Diaphoresis

Comps of cast Cont'd...

- Thrombophlebitis and possible pulmonary emboli ass'ted with immobility and ineffective circulation e.g. venous stasis
- Respiratory atelectasis and pneumonia ass'ted with ineffective respiratory effort
- UTI- renal and bladder calculi ass'ted with urinary stasis and low fluid intake and immobility
- Anorexia associated with constipation due to decreased activity
- Psychological rxn e.g. depressions associated with immobility, dependence and loss of control.

Nursing assessment 1st 24hrs of Cast

- ***Assess neurovascular status of the extremity with a cast for sign of compromise:***
 - Pain, Swelling
 - Discoloration, pale/ blue
 - Cool skin distal to injury
 - Tingling/ numbness-parasthesia
 - Pain on passive extension muscle stretch/
 - Slow capillary refill diminished or absent pulse
 - Paralysis

Nsg Ass't Cont'd...

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Assess skin integrity of casted extremity, be alert for:

- Severe initial pain over bony prominences this is a warning symptom of an impending pressure sore
- Pain increases when ulceration occurs

□ Carefully assess for positioning and potential pressure sites of the distal extremity:

- **Lower extremity:** heel, dorsum of foot, head of fibula, anterior surface of patella
- **Upper extremity:** medial epicondyle of humerus, ulnar styloid
- If **plaster jacket** or **body spica casts:** sacrum, anterior and superior iliac spine, vertebral borders of scapulae.

- Assess- CVS, RS, GIT for possible complications of immobility
- Assess- psychological rxn to illness, cast and immobility.
- Avoid resting cast on hard surface or sharp edges that can cause denaturing or flattening of cast and pressure sore
- Handle moist cast with palms of hands
- Turn pt Q2hrs while cast dry

- If symptoms of pressure are occurs cast may be “windowed” (hole cut in it) so the skin at a pain point can be examined and treated
- To avoid muscle atrophy and loss of strength isometric exercise for the muscles of the casted extremity. Instruct pt to alternatively contract and relax muscles moving the affected part.
- Use antiembolism/anticoagulant as prescribed.
- Encourage deep breathing exercise and coughing at regular intervals to prevent atelectasis and pneumonia

Observe for symptoms of cast syndrome

- **Cast syndrome** is a condition occurs if the body cast compresses the superior mesenteric artery against the duodenum.
- It is a rare sequel of body cast application
- It is potentially fatal complication
- It can develop as late as several weeks after cast application
- Encourage pt to drink liberal quantities of fluid to avoid urinary infection and calculi 2^o to immobility.

CAST/POP CARE

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DO NOT:

- ❑ Get plaster wet causes skin break down
- ❑ Remove any padding
- ❑ Insert any foreign object inside cast prevent pressure and injury to the skin
- ❑ Bear weight on new cast for 48 hrs (not all cast are made for weight bearing)
- ❑ Cover cast with plastic for prolonged periods it cause condensation and wetting of the cast.

Cast care Cont'd...

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DO

- Apply ice directly over # site for first 24 hrs (avoid getting cast wet by keeping ice in plastic bag and protecting cast with cloth)
- Dry cast thoroughly after exposure to water
- Elevate extremity above heart level for first 48hrs.
- Move joints above and below cast regularly
- **Report signs of possible problems like:**
 - Swelling associated with pain -- Increasing pain
 - Discoloration of toes/fingers -- Pain during movement
 - Burning or tingling under cast or
 - Sore or foul odor under cast
- Keep appointment to have # and cast check

Application of cast

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□ Equipment

- Non – absorbent cotton bandage and stockinet (if available) to protect the skin
- Absorbent cotton bandage to pad the bony prominences
- Plaster of paris (POP) bandage of appropriate size
- Rubber gloves gown and rubber sheet
- Basin or pail full of tepid water at a temp. of 21-24 °C
- Fracture knife and scissors
- Measuring tape, ruler and marker.

Procedure of cast application

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- Explain procedure to patient
- Wash the area very thoroughly and dry
- Apply non- absorbent cotton over the area for protection
- Apply tubular stockinet as lining for the cast, if available.
- Apply D.D.T powder over the cotton if needed
- Immerse the plaster bandage in the basin of warm water for 5 second and remove excess water carefully by applying slight pressure on it
- Apply wet plaster over the area leaving a portion of the stockinet at the ends to be folded back over the cast

N.B. -- When lifting the wet cast use the palms of the hands to avoid denoting into it.

-- Plaster castes are to not applied as tightly as other bandages b/c they shrink as they dry.

Care of the patient in cast

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□ Purpose

- To promote comfort , prevent pressure sores and other complication as deformity, thrombosis embolism constipation, etc
- To improve general health of the patient and speed up recovery.

□ Equipment

- Fracture board of the appropriate size
- Firm mattress and bed cradle
- Sand bags, 2-3
- Pillow 2-3
- A bed with trapez if available
- Macintosh

Cast care cont'd..

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i. Immediate care

- The bed should be prepared be fore-hand with a board under the mattress if needed
- Immediately after the completion of case application, the end parts of the extremity on which the cast is applied to move should be observed for symptoms such as pain, pallor discoloration, tingling sensation, numbness, coldness or inability
- Report immediately if any of these sings are noticed

Cast care cont'd..

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- Put mackintosh under the part where the cast is to protect the bed linen away from the wet plaster until it dries put the bed cradle to lift the weight off the bed linen
- Be informed about the condition of the patient and the type of cast applied
- If cast is applied around the trunk or hip, turn patient frequently to allow even drying of the cast and to prevent fatigue of the patient
- Elevation of the part may help to control swelling
- Heat should never be applied to the cast in attempt to dry it quickly as it can crack and cause burning also.

Cast care cont'd..

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ii. Care after cast is dry:

- A care includes protection of the skin and care of the cast
 - Pressure sores may developed over any bony prominence and commonly at the buttocks.
 - Wash and give pressure area care and change position frequently
 - Smoothen the rough edges and fix it with adhesive plaster.
 - Remove plaster crumbs from inside the cast with your hand massage the inside of the cast as far as you can reach.
 - Protect the cast around the perineal area against excretion by water proof material when the cast is dried.

Cast care cont'd..

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- The skin below the cast should be inspected frequently for swelling, discoloration, tingling, numbness, inability to move fingers or toes or temperature changes
- Prevent atrophy and maintain muscle strength by helping him to contract his muscles without moving the injured part. He should exercise unaffected part of the body regularly to develop his muscle tone and improve his general health.
- Elevating the head of the bed slightly when a bed pan is used helps to prevent soiling of the bed.
- If a cast is applied to the arm the patient should wear a sling to support the full weight of the arm and the hand must be supported to prevent development of wrist drop.

Removal of cast

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- **Cast is removed when: fracture is healed or cast is too tight**

- **Equipment**

- Electrical shears
- Metal stripe, if available
- Water and soap
- Towel
- Towel
- Oil or lanolin
- Waste basket
- Dressing scissors
- Plaster
- Knife

Procedure of Cast removal

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- Explain procedure to patient
- Make a mark with a plaster knife
- Use either the electric shears the plaster shears cutter and push it gently in to the cast
- Following the knife mark cut the last layer with dressing scissors
- Wash the skin with water and soap
- Dry it with a towel and apply oil or lanoline
- Discard the removed cast
- Remove, clean and return used equipment to its proper place
- Record.

Cast removal....

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REMEMBER:

- Be careful not to cut skin under the case while cutting cast
- Electric shears should be used by skilled personnel only
- Help the patient to understand that he may experience pain and stiffness until he is rehabilitated

A black and white photograph of two hands shaking, symbolizing agreement or integrity. The hands are positioned in the upper right quadrant of the slide, with the left hand (lighter skin) on top and the right hand (darker skin) on the bottom, fingers interlaced.

**Integrity
is doing the
right thing, even when
no one is watching.**

C.S. Lewis

Spirit Science